Aiden Zhang

437-788-7616 | a522zhan@uwaterloo.ca | linkedin.com/in/aidenzhangg | github.com/aidenzhangg

Technical Skills

Languages: Java, Python, C++, SQL, HTML, CSS, TypeScript, JavaScript
Frameworks: React.js, PyTorch, TensorFlow, scikit-learn, OpenCV, YOLO, ROS 2, pandas, numpy, Spring Boot
Developer Tools: Git, Github, Docker, VSCode, Foxglove, PyCharm, Eclipse, Vite, Chrome DevTools
Libraries: OpenAPI

Projects

Computer Vision Food Recognition and Nutritional Analysis System

Python, PyTorch, YOLO, OpenCV, NumPy, pandas, TensorFlow June 2025 - October 2026

- Built a real-time food detection app that finds and identifies 5 food categories in live webcam video, replacing an earlier TensorFlow classifier (96% test accuracy) with a fine-tuned YOLO11 model that boxes each food and detects several items at once averaging ~13.5 FPS across 300 frames
- Trained the model on **245 hand-labeled images, reaching 98% mAP50** on a held out validation set
- Ran a 5-month pilot with **6 peers and family members,** including a **visually impaired participant** who used TTS to access nutrition feedback supporting nutrition-aware food choices; participants reported healthier eating habits and an average **1.1% reduction in body-fat percentage**

Autonomous Robot Navigation & LiDAR Mapping - WATonomous

C++, ROS2, Foxglove, Docker Sept 2026

- Developed a 4-node autonomous navigation pipeline in C++ and ROS 2, integrating LiDAR mapping, odometry-based map fusion, A* pathfinding, and 10Hz Pure Pursuit control
- Engineered 100 x 100 m occupancy grids with 0.1m resolution, 0.65 m obstacle inflation, and 8-direction A* planning for obstacle-aware navigation

Autonomous Drone Behaviour Tree Telemetry System

Python, ROS2, MAVLINK, py_trees, pymavLink, Git Sept 2026 - Present

- Implemented an initial ROS 2 behaviour-tree telemetry protocol, subscribing to live airside snapshots and automatically mapping 100% of observed behaviors to compact IDs and parent-child relationships for bandwidth-constrained MAVLink STATUSTEXT
- Developed an air-to-ground telemetry pipeline to encode active decision paths and reconstruct autonomy state groundside, targeting ≤ 50-byte STATUSTEXT payloads for debugging without LTE connectivity

Experience

Camp Counsellor - *City of Richmond Hill* May 2025 - August 2026

- Led daily programs for children ages 4-12, working with staff and parents to resolve issues

Certificates

Stanford AI Machine Learning Specialization July 2025 - October 2025

PyCharm, PyTorch, Python, TensorFlow, Scikit-learn, NumPy, Langchain, LLM

- Built ML models using neural networks, regression, decision trees, clustering, and recommender systems

Education

University of Waterloo - *BASc in Computer Engineering with Co-op* Sept 2026 - April 2031

- **Coursework:** C++ Programming, Physics, Linear Algebra, Calculus, Engineering Prototype and Design

Aiden Zhang

437-788-7616 | a522zhan@uwaterloo.ca | linkedin.com/in/aidenzhangg | github.com/aidenzhangg

Education

University of Waterloo - *BASc in Computer Engineering with Co-op* Sept 2026 - April 2031

- **Relevant Coursework:** C++ Programming & Object-Oriented Design, Algorithms & Data Structures, Engineering Design & Prototyping

Technical Skills

Languages: Java, Python, C++, SQL, HTML/CSS
Frameworks: React.js, TensorFlow, Scikit-learn, Langchain, PyTest, ROS 2
Developer Tools: Git, Github, VS Code, Eclipse, PyCharm, Docker
Libraries: PyTorch, OpenCV, pandas, NumPy

Projects

Computer Vision Food Recognition and Nutritional Analysis System

Python, React.js, OpenCV, NumPy, pandas, TensorFlow Dec 2024 - June 2026

- Developed and tested a real-time AI food recognition system using TensorFlow and OpenCV, classifying 5 food categories from live webcam input with **96%** test accuracy
- Coordinated and communicated pilot trials through the system with **6 peers and family members over 5 months,** including a visually impaired participant who used TTS to access nutrition feedback supporting nutrition-aware food choices; participants reported healthier eating habits and an average **1.1% reduction in body-fat percentage**
- Fine-tuned a convolutional neural network (CNN) with image preprocessing and integrated a nutrition database and text-to-speech (TTS) for real-time macronutrients, health, and auditory feedback

Autonomous Drone Behaviour Tree Telemetry System

Python, ROS 2, MAVLINK, py_trees, pymavLink, Git Sept 2026 - Present

- Implemented an initial ROS 2 behaviour-tree telemetry protocol, subscribing to live airside snapshots and automatically mapping 100% of observed behaviors to compact IDs and parent-child relationships for bandwidth-constrained MAVLink STATUSTEXT
- Developed an air-to-ground telemetry pipeline to encode active decision paths and reconstruct autonomy state groundside, targeting ≤ 50-byte STATUSTEXT payloads for debugging without LTE connectivity

Autonomous Drone Software & Computer Vision Bootcamp - WARG

Python, NumPy, PyTest, Git, YAML, CI/CD Sept 2026

- Engineered a simulated camera sensor in Python/NumPy implementing an abstract camera interface, predetermined 3-channel uint8 frame generation
- Developed a comprehensive PyTest suite for autonomous waypoint utilities while achieving 90%+ branch coverage and eliminating 12/12 mutants through mutant testing

Certificates

Stanford AI Machine Learning Specialization July 2025 - October 2025

PyCharm, PyTorch, Python, TensorFlow, Scikit-learn, NumPy, Langchain, LLM

- Built ML models using neural networks, regression, decision trees, clustering, and recommender systems

Experience

Camp Counsellor - *City of Richmond Hill* May 2025 - August 2026

- Coordinated daily programs for children ages 4-12, maintaining schedules and activities
- Collaborated with staff, supervisors, and parents to resolve issues and adapt to changing needs, demonstrating leadership, communication, teamwork, and time management

Recall - AI Study Assistant Chrome Extension

Java 21, Spring Boot, React, TypeScript, PostgreSQL, SQL, Vite, CSS, Git, npm, Vite October 2026 - Present

- Built a chrome Manifest V3 side-panel extension with React and TypeScript that generates fill-in-the-blank quizzes from pasted or uploaded course notes with Hints On/Off modes & retry of missed questions
- Designed a Java 21 and Spring Boot REST API backed by PostgreSQL that persists quizzes, questions and attempts across sessions

Junior Developer - Intern | MJVA Technology Ltd. Richmond Hill, ON

June 2026 - September 2026

Aiden Zhang

a522zhan@uwaterloo.ca
linkedin.com/in/aidenzhangg
github.com/aidenzhangg
437-788-7616

Computer Engineering at the University of Waterloo

Computer Vision Food Recognition & Nutritional Analysis - Personal Project
WHAT:

HOW:

RESULT:

Autonomous Robot Navigation & LiDAR Mapping — WATonomous
WHAT:

HOW:

RESULT:

Recall: Study Extension - Personal Project
WHAT:

HOW:

RESULT:

Smart Patient Health & Motion Monitoring System - Personal Project
Arduino UNO, C++, Embedded Programming, I^2C, UART, Serial Communication, ADC, Signal Processing, Git, Github, Arduino IDE
WHAT:

HOW:

RESULT:

Autonomous Drone Behaviour Tree Telemetry — WARG
WHAT:

HOW:

RESULT:

Drone Emergency Abort System — WARG
WHAT:

HOW:

RESULT:

- Add ece 198 to resume → include both imu sensor -> tracking exercise

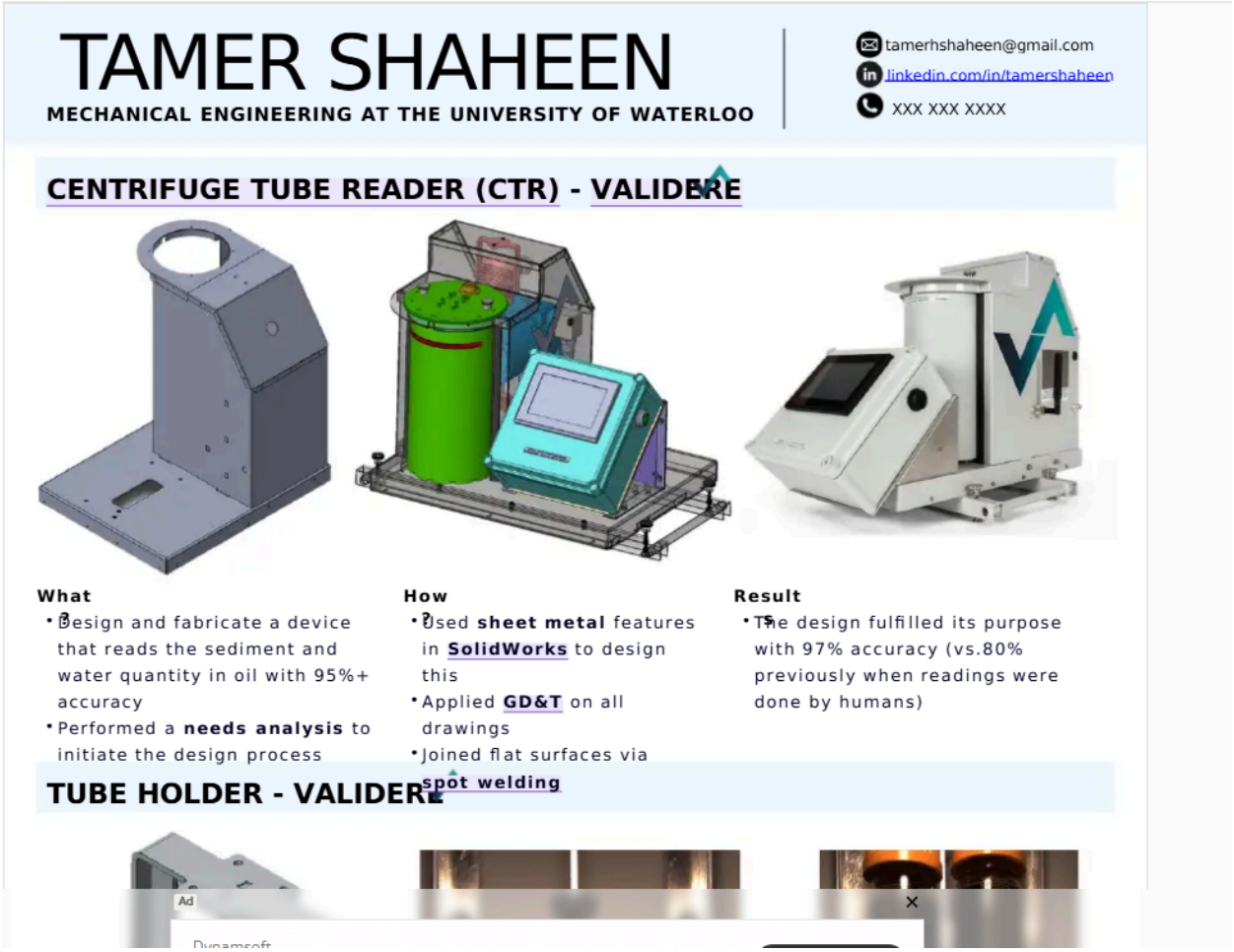
CCCARE Exercise Tracking & Rehabilitation Monitoring System

- Developed IMU
- 1:30

Junior developer - MJVA Technology Ltd.

WARG Airside Safety supervisor → abort

- Built an emergency abort for an autonomous drone's ROS 2 behavior tree (Python, py_trees). A safety check runs on every tick and, when the ground station sends ABORT, stops the active mission and commands an immediate landing through MAVROS.



Computer Vision